

AstraX – Major EB1 Engineering Risks

Engineering Management Review

Ametek Tube Part Impact

Risk: Single-source critical component

Impact: Delays block radiation validation & calibration

Mitigation: Escalation, interim emulator, alternate sourcing study

PDLC Discipline Gap

Risk: Requirements fragmented across Teams, Excel, drawings

Impact: No RTM, ICD, or formal baselines

Interface Risks – HW↔FW↔SW↔Cloud↔Mech

Risk: Voltage, timing, thermal, protocol mismatch

Vendor Sourcing & Contracts

Risk: Weak milestone enforcement

Impact: Incentives not tied to EB1 validation

Mitigation: Milestone-based payments + delay clauses

EE Risks

Risk: ERS review gaps, USB/Battery FW complexity, distributed team

Impact: Underestimated validation duration, bring-up risk

Mitigation: Deep technical review + integration war-room

Long Lead Items – Jig & EB1

Risk: CNC parts, detector housing, PCB fab queue

Impact: Jig +1 mo, EB1 +2 mo schedule drift

Mitigation: BOM freeze, parallel vendor validation

CalcEngine Dependency

Risk: Core spectral computation reliance

Impact: IP/licensing clarity + performance portability risk

Risk: Schedule compression impacts burn runway

Impact: Rework probability increasing

Mitigation: Strict change control + weekly burn review

Mechanical Risks

Risk: Drop validation, IP54 sealing, radiation safety & enclosure testing

Impact: Compliance failure risk at system integration

Mitigation: Pre-EB1 mechanical stress validation plan

SW / FW Risks

Risk: HW delays pushing XRF testing, LLM tool dependency

Impact: Small-screen UX constraints, reliability/control concerns

Mitigation: Decouple test layers + fallback computation paths

EB1 success depends on integration discipline, supplier control, and artifact maturity.